Day 2 - SQL Basics Teaching Guide (60 Minutes)

Class Schedule

Time	Topic
0-5 min	Recap Day 1
5-15 min	SELECT Statement
15-30 min	WHERE Clause
30-40 min	ORDER BY
40-50 min	DISTINCT
50-58 min	TOP (SQL Server) / LIMIT
58-60 min	Practice & Doubts

Step 1 - Recap

Ask students: What is a Database? Table? Row? Column? Which command creates a table? Then explain that today they will learn how to retrieve data.

Sample Dataset

```
CREATE TABLE Employee(
EmployeeID INT,
EmployeeName VARCHAR(50),
Department VARCHAR(50),
Salary INT,
City VARCHAR(50));

INSERT INTO Employee VALUES
(101,'Rishi','IT',60000,'Chennai'),
(102,'Priya','HR',50000,'Bangalore'),
(103,'Rahul','IT',70000,'Chennai'),
(104,'Anu','Finance',55000,'Hyderabad'),
(105,'John','IT',70000,'Chennai');
```

Topic 1 - SELECT

Start by asking: 'How do we see data from a table?'
Explain:
SELECT * FROM Employee;
Show selecting specific columns:

SELECT EmployeeName, Salary FROM Employee;
Practice: Display EmployeeID and City.

Topic 2 - WHERE

Explain filtering:
SELECT * FROM Employee WHERE Department='IT';
Examples:
Salary > 60000
City='Chennai'
BETWEEN
AND
Practice: Show employees from Bangalore.

Topic 3 - ORDER BY

Sorting:
SELECT * FROM Employee ORDER BY Salary;
SELECT * FROM Employee ORDER BY Salary DESC;
Practice: Sort by City.

Topic 4 - DISTINCT

Explain duplicate removal:
SELECT DISTINCT Department FROM Employee;
SELECT DISTINCT City FROM Employee;
Practice: Find unique salaries.

Topic 5 - TOP

SQL Server:
SELECT TOP 3 * FROM Employee;
Highest paid:
SELECT TOP 1 * FROM Employee ORDER BY Salary DESC;
Mention MySQL uses LIMIT.

Combined Example

SELECT TOP 2 *
FROM Employee
WHERE Department='IT'
ORDER BY Salary DESC;

Explain execution order:

1. FROM
2. WHERE
3. SELECT
4. ORDER BY
5. TOP

Practice Questions

1. Display all employees.
2. Display EmployeeName and Salary.
3. Show employees from Chennai.
4. Salary > 55000.
5. Employees in IT.
6. Sort Salary ASC.
7. Sort Salary DESC.
8. Unique Departments.
9. Unique Cities.
10. Top 3 highest-paid employees.

Key Summary

SELECT - Which columns?
FROM - Which table?
WHERE - Which rows?
ORDER BY - Sorting
DISTINCT - Unique values
TOP - First N rows